

Section A: Git Basics + Setup

Q1.Check if Git is installed. What version do you have? Use Command

```
om@om-HP-Laptop-15s-fq4xxx:~$ git --version
git version 2.53.0
om@om-HP-Laptop-15s-fq4xxx:~$ git config --global user.name omClover
om@om-HP-Laptop-15s-fq4xxx:~$ git config --global user.email om.chaudhari@cloverinfotech.com
```

Q2.Set your name and email in global config.

```
om@om-HP-Laptop-15s-fq4xxx:~$ mkdir git_assignment_om
om@om-HP-Laptop-15s-fq4xxx:~$ cd git_assignment_om/
om@om-HP-Laptop-15s-fq4xxx:~/git_assignment_om$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /home/om/git_assignment_om/.git/
```

Q3.Create a new folder `git-assignment-YourName` and run `git init` inside it. Check the status.

```
om@om-HP-Laptop-15s-fq4xxx:~$ mkdir git_assignment_om
om@om-HP-Laptop-15s-fq4xxx:~$ cd git_assignment_om/
om@om-HP-Laptop-15s-fq4xxx:~/git_assignment_om$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /home/om/git_assignment_om/.git/
om@om-HP-Laptop-15s-fq4xxx:~/git_assignment_om$ touch asnswers.md
```

Q4.Create a `README.md` file with content `# Git Assignment - YourName`. Stage it and commit with message "Initial commit".

```
om@om-HP-Laptop-15s-fq4xxx:~/git_assignment_om$ cat > README.md
Git Assignment-0m
om@om-HP-Laptop-15s-fq4xxx:~/git_assignment_om$ git add .
om@om-HP-Laptop-15s-fq4xxx:~/git_assignment_om$ git commit -m "Initial Commit"
[master (root-commit) 1a24ccd] Initial Commit
 2 files changed, 1 insertion(+)
 create mode 100644 README.md
 create mode 100644 asnswers.md
```

Q5.Run `git log` and copy the hash ID of your first commit into `answers.md`.

1a24ccdda45d243fd050c0879847d2548888d719

Q6.Perform few changes and make a five commits.

This is change one

This is change two

This is change three

This is change four

This is change five

```
om@om-HP-Laptop-15s-fq4xxx:~/git_assignment_om$ git commit -am "second Commit"
[master 32887df] second Commit
 1 file changed, 118 insertions(+)
om@om-HP-Laptop-15s-fq4xxx:~/git_assignment_om$ git commit -am "second Commit"
[master 6cffda6] second Commit
 1 file changed, 3 insertions(+), 1 deletion(-)
om@om-HP-Laptop-15s-fq4xxx:~/git_assignment_om$ git commit -am "third Commit"
[master a5e968e] third Commit
 1 file changed, 2 insertions(+)
om@om-HP-Laptop-15s-fq4xxx:~/git_assignment_om$ git commit -am "fourth Commit"
[master e8e8a86] fourth Commit
 1 file changed, 2 insertions(+)
om@om-HP-Laptop-15s-fq4xxx:~/git_assignment_om$ git commit -am "fifth Commit"
[master ce0212a] fifth Commit
 1 file changed, 2 insertions(+)
```

Section B: Branching & Commits

Q6. Create a new branch named `dev` and switch to it. Show the branch list.

```
om@om-HP-Laptop-15s-fq4xxx:~/git_assignment_om$ git checkout -b dev
Switched to a new branch 'dev'
om@om-HP-Laptop-15s-fq4xxx:~/git_assignment_om$ git branch
* dev
master
```

Q7. On the `dev` branch, create `features.txt`. Add 3 lines: "Login", "Signup", "Dashboard". Commit the changes.

```
om@om-HP-Laptop-15s-fq4xxx:~/git_assignment_om$ git checkout -b dev
Switched to a new branch 'dev'
om@om-HP-Laptop-15s-fq4xxx:~/git_assignment_om$ git branch
* dev
master
om@om-HP-Laptop-15s-fq4xxx:~/git_assignment_om$ cat features.txt
cat: features.txt: No such file or directory
om@om-HP-Laptop-15s-fq4xxx:~/git_assignment_om$ cat > features.txt
Login
Signup
Dashboard
om@om-HP-Laptop-15s-fq4xxx:~/git_assignment_om$ git add .
om@om-HP-Laptop-15s-fq4xxx:~/git_assignment_om$ git commit -m "Added feature.txt"
[dev 0782787] Added feature.txt
2 files changed, 4 insertions(+)
create mode 100644 .~lock.asnswers.md#
create mode 100644 features.txt
```

Q8. Switch back to `main`/`master` branch. Is `features.txt` visible? It should not be. Write the reason.

```
om@om-HP-Laptop-15s-fq4xxx:~/git_assignment_om$ git checkout master
Switched to branch 'master'
om@om-HP-Laptop-15s-fq4xxx:~/git_assignment_om$ ls
README.md  asnswers.md
```

The file is not visible in the branch 'master' because it only present in the snapshot of 'dev' branch and doesn't merged into 'master' branch

Q9. Merge `dev` into `main`. After merging, take a screenshot of `git log --oneline`.

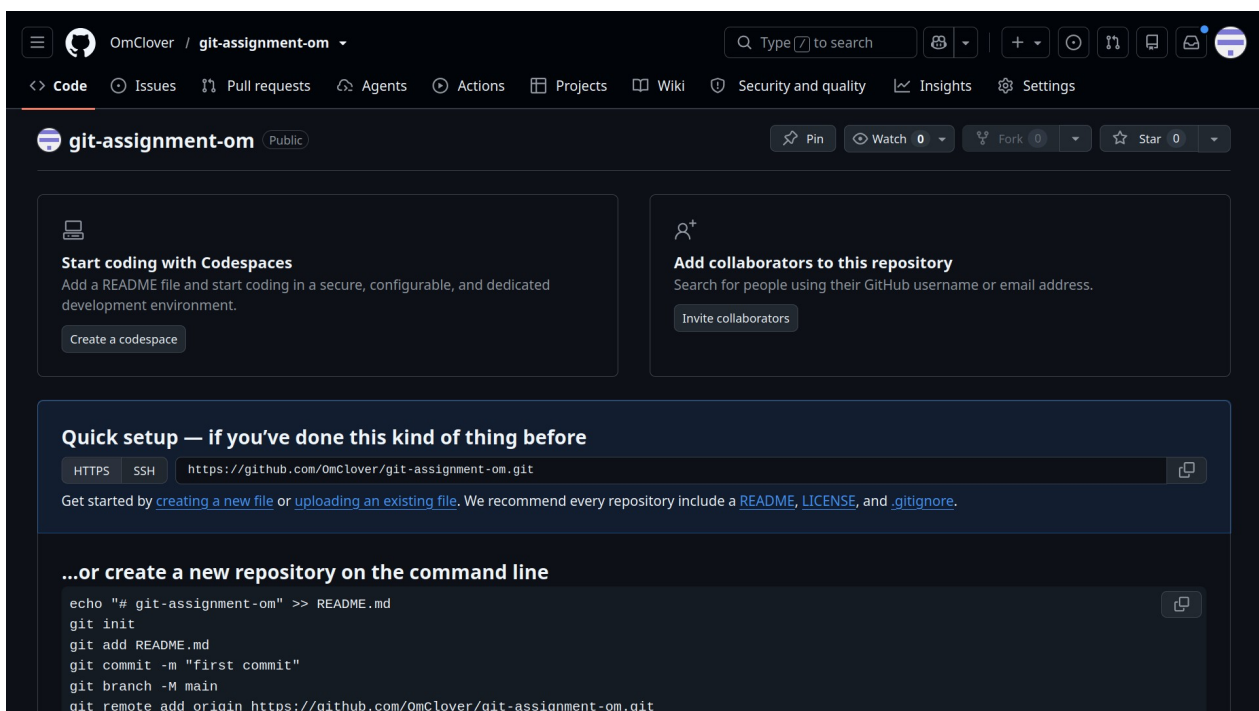
```
om@om-HP-Laptop-15s-fq4xxx:~/git_assignment_om$ git log --oneline
0782787 (HEAD -> master, dev) Added feature.txt
ce0212a fifth Commit
e8e8a86 fourth Commit
a5e968e third Commit
6cffda6 second Commit
32887df second Commit
1a24ccd Initial Commit
```

Q10. Create a `.gitignore` file. Add `node\_modules/` and `log` to it. Commit the file.

```
om@om-HP-Laptop-15s-fq4xxx:~/git_assignment_om$ mkdir .gitignore
om@om-HP-Laptop-15s-fq4xxx:~/git_assignment_om$ touch log
om@om-HP-Laptop-15s-fq4xxx:~/git_assignment_om$ mkdir node_modules
om@om-HP-Laptop-15s-fq4xxx:~/git_assignment_om$ git add .
om@om-HP-Laptop-15s-fq4xxx:~/git_assignment_om$ git commit -m "Added .gitignore"
[master 0325321] Added .gitignore
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 log
```

Section C: GitHub Remote + Push/Pull

Q11. Create a new empty repo on GitHub named `git-assignment-YourName`. Do not initialize with README.



The screenshot shows the GitHub interface for a repository named 'git-assignment-om' by user 'OmClover'. The repository is public and has 0 watches, 0 forks, and 0 stars. The page includes a 'Quick setup' section with instructions for cloning the repository using HTTPS or SSH, and a section for creating a new repository on the command line. The command line instructions are as follows:

```
echo "# git-assignment-om" >> README.md
git init
git add README.md
git commit -m "first commit"
git branch -M main
git remote add origin https://github.com/OmClover/git-assignment-om.git
```

Q12. Connect your local repo to the GitHub remote.

```
git remote add origin https://github.com/OmClover/git-assignment-om.git
```

Q13. Push your local code to GitHub.

```
● om@om-HP-Laptop-15s-fq4xxx:~/git_assignment_om$ git push -u origin master
Enumerating objects: 25, done.
Counting objects: 100% (25/25), done.
Delta compression using up to 8 threads
Compressing objects: 100% (22/22), done.
Writing objects: 100% (25/25), 35.21 KiB | 11.74 MiB/s, done.
Total 25 (delta 5), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (5/5), done.
To https://github.com/OmClover/git-assignment-om.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.
```

Q14. Go to GitHub and paste your repo link in.

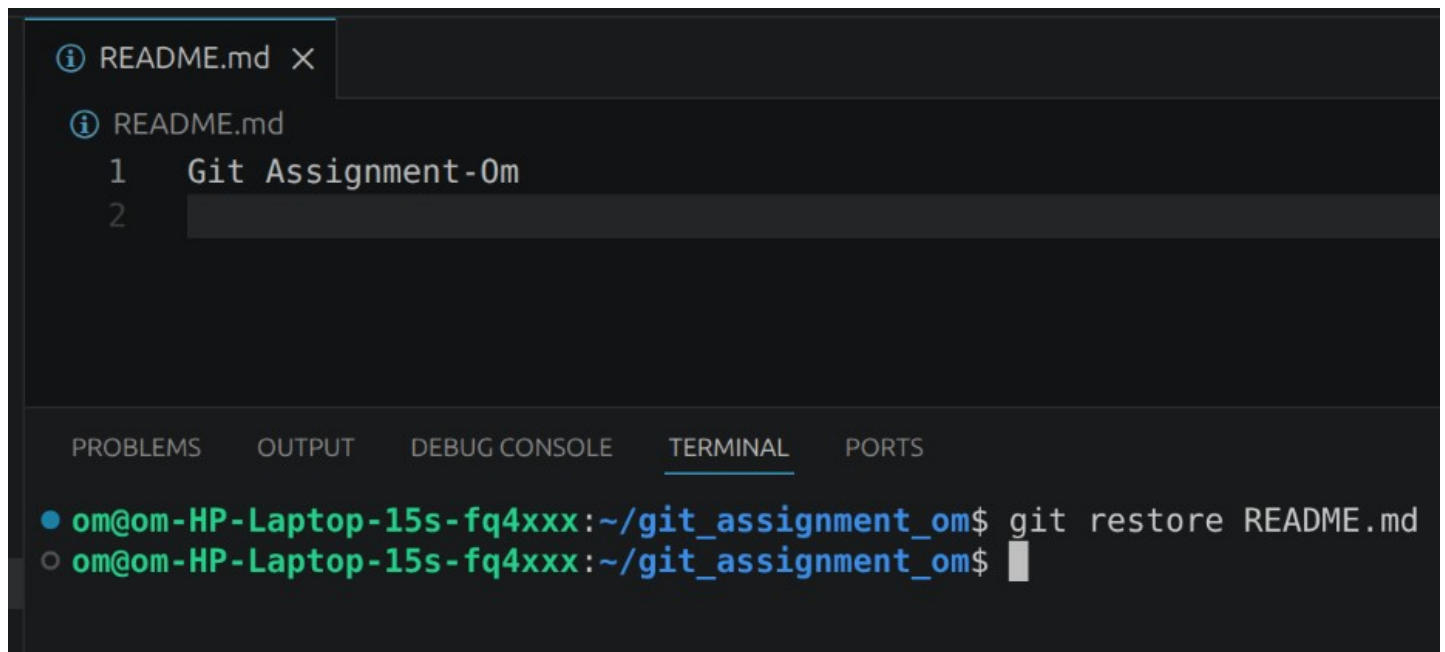
```
https://github.com/OmClover/git-assignment-om.git
```

Q15. Create a file directly on GitHub named `bugfix.txt` with content "Bug 1 fixed". Now run `git pull` locally to fetch that file.

```
● om@om-HP-Laptop-15s-fq4xxx:~/git_assignment_om$ git pull origin master
remote: Enumerating objects: 4, done.
remote: Counting objects: 100% (4/4), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 929 bytes | 929.00 KiB/s, done.
From https://github.com/OmClover/git-assignment-om
 * branch            master      -> FETCH_HEAD
    0325321..b094b08  master      -> origin/master
Updating 0325321..b094b08
Fast-forward
 bugfix.txt | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 bugfix.txt
```


Section D: Undo, Conflicts & Collaboration

Q16. Add a line "Testing undo" to `README.md`. Do not stage it. Now discard the changes using `git restore README.md`. Verify it worked.



The screenshot shows the VS Code editor with a file named `README.md` open. The file content is:

```
1 Git Assignment-0m
2
```

Below the editor, the `TERMINAL` tab is active, showing the following commands and output:

```
om@om-HP-Laptop-15s-fq4xxx:~/git_assignment_om$ git restore README.md
om@om-HP-Laptop-15s-fq4xxx:~/git_assignment_om$
```

Q17. The last commit message was wrong. Change it to "Initial commit - Added README".

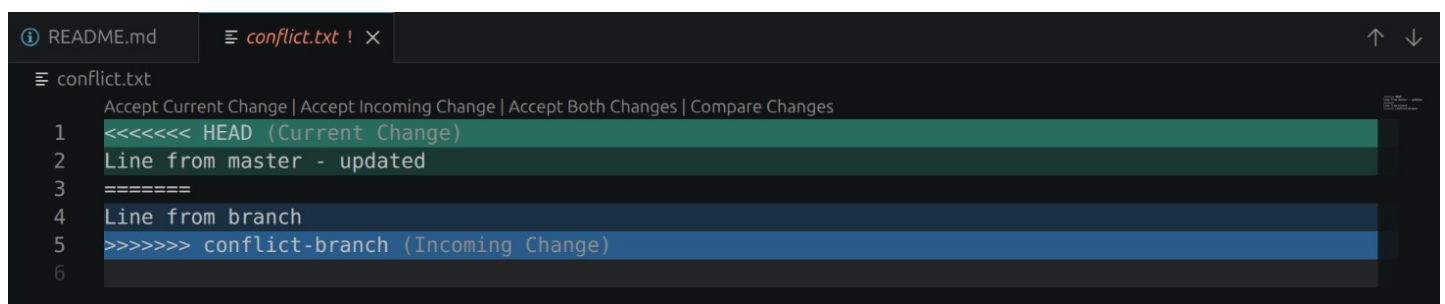
Hint: `git commit --amend`



```
om@om-HP-Laptop-15s-fq4xxx:~/git_assignment_om$ git commit --amend -m "Initial commit - Added README"
[master 6817916] Initial commit - Added README
Date: Sat Jun 27 21:01:26 2026 +0530
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 screenshots/Q16.png
```

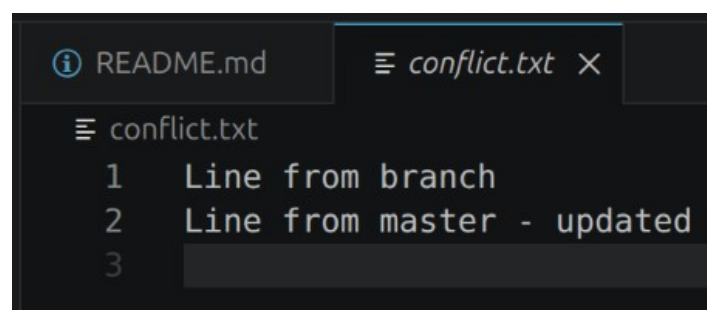
Q18. Conflict Task:

1. On `main` branch, create `conflict.txt` with "Line from main" and commit
2. Create a new branch `conflict-branch`, change the same file to "Line from branch" and commit
3. Try to merge into `main` - you will get a conflict
4. Manually resolve the conflict, keep both lines, and commit. Show the final content of `conflict.txt`.



The screenshot shows the VS Code editor with a file named `conflict.txt` open. The file content is:

```
1 <<<<<< HEAD (Current Change)
2 Line from master - updated
3 =====
4 Line from branch
5 >>>>>> conflict-branch (Incoming Change)
6
```



The screenshot shows the VS Code editor with a file named `conflict.txt` open. The file content is:

```
1 Line from branch
2 Line from master - updated
3
```

Q19. Fork a colleague's repo and clone it locally. Write the repo name in `answers.md`.

Repository Name: git-assignment-vivek.git

```
om@om-HP-Laptop-15s-fq4xxx:~$ git clone https://github.com/OmClover/git-assignment-vivek.git
Cloning into 'git-assignment-vivek'...
remote: Enumerating objects: 27, done.
remote: Counting objects: 100% (27/27), done.
remote: Compressing objects: 100% (13/13), done.
remote: Total 27 (delta 2), reused 24 (delta 1), pack-reused 0 (from 0)
Receiving objects: 100% (27/27), done.
Resolving deltas: 100% (2/2), done.
```

Q20. Add a collaborator to your `git-assignment-YourName` repo. Settings > Collaborators > Add people. Take a screenshot.

